Distance Rules Do Not Protect Atolls

Policy brief from DOUBLE JEOPARDY · Version 2.1 · Sakshi D. Maske

Key points

- On the Maldives, Lakshadweep, Seychelles and Fiji, 92–99.8% of buildings standing at or below 1 m are outside every statutory coastal setback. Half of these low buildings stand 80–136 m from the shoreline.
- A wider setback would not fix this on atolls. Spain's 100 m rule applied to the Maldives would reach 54% of low buildings but would also cover 39% of all buildings, because on atolls low land is most of the island, not a coastal strip.
- Having more coastal rules is not the same as reaching the buildings at risk. Lakshadweep has five of the six coastal instrument types coded in this study and its 20 m zone reaches 7% of its low buildings; the Maldives has two of the six and no setback, and its low buildings are almost all outside any rule too.
- The one place where a setback works is the Canary Islands, where low buildings form a narrow beachfront strip and Spain's 100 m line reaches 85% of them.
- Rules can also move backwards. Spain relaxed its coastal law in 2013, and the Maldives widened exemptions from environmental review in 2024–2025.

The problem

Coastal setbacks, a minimum distance between buildings and the shoreline, are the most common rule governments use to keep new buildings out of harm's way. They assume that the land most at risk from the sea is the land closest to it. On high islands and continental coasts that is roughly true. On coral atolls it is not: the whole island is low, and the buildings at risk stand in the middle of it as much as at its edge.

What the study measured

DOUBLE JEOPARDY compared five island groups governed by five different governments: the Maldives, Lakshadweep (India), Seychelles, Fiji and the Canary Islands (Spain). For about 1.35 million building footprints it measured each building's height on a bare-earth elevation model and its distance to the shoreline, and checked whether buildings at or below 1 m fall inside the setback written in national law. It also coded each government's coastal rules in force in October 2026 and 23 dated changes to those rules since 1988.

	Maldives	**Lakshadweep**	**Seychelles**	**Fiji**	**Canary Islands**
Statutory setback on inhabited land	none	20 m	25 m	none	20–100 m
Coastal instrument types in force (of 6)	2	5	3	3	6
Low buildings reached by the setback	0%	6.8%	8.0%	0%	84.9%
Setback needed to reach 75% of low buildings	151 m	187 m	262 m	262 m	62 m

Recommendations

1. **On atolls and low islands, regulate by height, not by distance.** Minimum floor levels or ground levels above a reference water level reach low buildings wherever they stand on the island. A distance rule can be kept for beach protection, but it should not be relied on to reduce flood exposure.
2. **Check rules against where buildings actually are.** Open building footprints and bare-earth elevation models now make it possible to test, for any island, what share of low buildings a proposed rule would reach. This test is cheap and can be repeated whenever a rule is revised.
3. **Use bare-earth elevation data.** On forested islands, surface elevation models read the tree canopy as ground and can understate the share of people on low land by five to eight times.
4. **Count instruments with care.** Comparisons of coastal policy that count laws, plans and agencies can rank a government highly even when its rules reach few of the people at risk. Outcome measures should sit next to output measures.
5. **Watch for quiet rollbacks.** Exemptions from impact assessment and extensions of existing coastal concessions weaken protection without repealing any rule. Tracking amendments over time shows these changes.

Limits

The shoreline is the edge of mapped island polygons, not the legal reference line, so distances are approximate by some tens of metres. Elevation has an error of about 0.4–0.5 m. The Lakshadweep setback width and the Maldivian changes of 2024–2025 rest on secondary sources. The analysis shows where buildings are, not whether any rule was broken.

Full study: Research Paper, Sections 3.9–3.10 and 4.8–4.9 · Code and data: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY · Zenodo DOI: 10.5281/zenodo.23827474